

Name:	Kaushik Balaji.M.S
Roll no:	152502010

CS5107 Programming Lab

Assignment 1

Question:

What if the number of arguments passed to a bash script are more than 9?

Method 1

Mentioning every arguments explicitly with their argument number. To mention arguments from 1-9, we use \$1, \$2, ... \$9. Beyond that, we mention the arguments using, \${10}, \${11}, ...



The screenshot shows a code editor with a file named `way1.sh`. The script content is as follows:

```
1  #!/bin/bash
2
3  #   Average of 15 input numbers
4
5  #   Method 1
6  #   access 1-9 args as $1, $2, ... . after that use ${n}, n>9
7
8  sum=$(( $1+$2+$3+$4+$5+$6+$7+$8+$9+${10}+${11}+${12}+${13}+${14}+${15} )) # have to mention every inputs this way
9  average=$(( $sum/15 ))
10
11  echo "$average"
12
13
14
```

Below the code editor is a terminal window. The first prompt shows the command `./way1.sh 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15` being executed. The output of the script is `8`. The second prompt shows the user at the shell.

In this, we have to mention every arguments (from 1-9 as \$1 ... \$9 and >10 as \${10} ...).

Method 2

```
$ way1.sh $ way2.sh x
$ way2.sh
1  #!/bin/bash
2
3  #   Average of N input numbers
4
5  #   Method 2
6  #   Get the total no of args with $# and access every arguments that are given using ${!i}
7
8  echo "Number of arguments is: $#"
```

```
9
10 sum=0
11 for(( i=1; i<=$#, i++ )); do
12 |   (( sum = sum + ${!i}))
13 done
14 average=$(( $sum/$# ))
15
16 echo "Average of given $# numbers: $average"
```

```
bash: ./way2.sh: Permission denied
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$ chmod +x way2.sh
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$ ./way2.sh 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Number of arguments is: 15
Average of given 15 numbers: 8
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$ ./way2.sh 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Number of arguments is: 20
Average of given 20 numbers: 10
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$
```

Here, we can get the total number of arguments passed to the bash script using `$#` which holds the value of no. of argument passed. Then we can iterate with a loop, and get the individual arguments using `${!i}`. This way even if we don't know the number of arguments, we can use every arguments mentioned.

Method 3

```
$ way1.sh $ way2.sh $ way3.sh x
$ way3.sh
1  #!/bin/bash
2
3  #   Average of N input numbers
4
5  #   Method 3
6  #   Get the list of arguments using $@ and iterate through them
7
8  echo "List of arguments is: $@"
9
10 sum=0
11 for num in "$@"; do
12 |   (( sum = sum + $num))
13 done
14 average=$(( $sum/$# ))
15
16 echo "Average of given $# numbers: $average"
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$ chmod +x way3.sh
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$ ./way3.sh 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
List of arguments is: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Average of given 20 numbers: 10
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$ ./way3.sh 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
List of arguments is: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Average of given 15 numbers: 8
● kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Assignment/Assignment 1$
```

Here, we use `$@` to get the list of all arguments given during the execution, and iterate through these arguments in a loop.

